① · In how many ways can the 26 letters of the English alphabet be permuted so that none of the Patterns CAR, DOG, PUN or BYTE occurs?

➤ Let S be the set of all arrangements of 26 letters. Then $|S| = N = S_0 = 26!$

Let C_1 denotes the Symbol CAR appears in the arrangement.
Let C_2 denotes the Symbol DOG appears in the arrangement.
Let C_3 denotes the Symbol PUN appears in the arrangement.
Let C_4 denotes the Symbol BYTE appears in the arrangement.

By the Principal of Inclusion - Exclusion, we have

$$\bar{N} = S_0 - S_1 + S_2 - S_3 + S_4$$

Here $S_1 = \sum N(C_i)$

$N(C_1) = [26 - 3 + 1]! = 24!$

$N(C_1)$ → The No. of arrangement of 26 letters such that the Symbol CAR appears.

$N(C_2) = [26 - 3 + 1]! = 24!$
$N(C_3) = [26 - 3 + 1]! = 24!$
$N(C_4) = [26 - 4 + 1]! = 23!$

$\therefore S_1 = N(C_1) + N(C_2) + N(C_3) + N(C_4) = 24! + 24! + 24! + 23!$

$= 3[24!] + 23!$

Since $S_2 = \sum N(C_i C_j) = N(C_1C_2) + N(C_1C_3) + N(C_1C_4) + N(C_2C_3) + N(C_2C_4) + N(C_3C_4)$

$N(C_1C_2) = [26 - 3 + 1 - 3 + 1]! = 22!$
$N(C_1C_3) = [26 - 3 + 1 - 3 + 1]! = 22!$
$N(C_1C_4) = [26 - 3 + 1 - 4 + 1]! = 21!$
$N(C_2C_3) = [26 - 3 + 1 - 3 + 1]! = 22!$
$N(C_2C_4) = [26 - 3 + 1 - 4 + 1]! = 21!$
$N(C_3C_4) = [26 - 3 + 1 - 4 + 1]! = 21!$

$\therefore S_2 = 3[22!] + 3[21!]$

To find S_3, we have $S_3 = \sum N(C_i C_j C_k)$

$\Rightarrow S_3 = N(C_1 C_2 C_3) + N(C_1 C_2 C_4) + N(C_2 C_3 C_4)$

$N(C_1 C_2 C_3) = [26 - 3 + 1 - 3 + 1 - 3 + 1]! = 20!$

$N[C_1 C_2 C_4] = [26 - 3 + 1 - 3 + 1 - 4 + 1]! = 19!$

$N[C_2 C_3 C_4] = [26 - 3 + 1 - 3 + 1 - 4 + 1]! = 19!$

$\therefore S_3 = 2[19!] + 20!$

To find S_4, we have $S_4 = \sum N(C_i C_j C_k C_l) = N(C_1 C_2 C_3 C_4)$

$N(C_1 C_2 C_3 C_4) = [26 - 3 + 1 - 3 + 1 - 3 + 1 - 4 + 1]! = 17!$

$\therefore \bar{N} = S_0 - S_1 + S_2 - S_3 + S_4$

$= 26! - [3(24!] + 23!] + [3(22!] + 3(21!)] - [2(19!) + 20!] + 17!$

(2). Find the number of permutations of the letters a, b, c, ---, x, y, z in which none of the patterns ~~spin~~ spin, game, path or net occurs.

▶ Let S be the set of all arrangements of 26 letters.

Then $|S| = N = S_0 = 26!$

Let C_1 denotes the Symbol spin appears in the arrangement.

Let C_2 denotes the Symbol game appears in the arrangement

Let C_3 denotes the Symbol path appears in the arrangement.

Let C_4 denotes the Symbol net appears in the arrangement.

By the principle of Inclusion - Exclusion, we have

$\bar{N} = S_0 - S_1 + S_2 - S_3 + S_4$

To find S_1, we have $S_1 = \sum N(C_i) = N(C_1) + N(C_2) + N(C_3) + N(C_4)$

$N(C_1) = [26 - 4 + 1]! = 23!$

$N(C_2) = [26 - 4 + 1]! = 23!$

$N(C_3) = [26 - 4 + 1]! = 23!$

$N(C_4) = [26 - 3 + 1]! = 24!$

$\therefore S_1 = 23! + 23! + 23! + 24!$

$= 3[23!] + 24!$

we have $S_2 = \sum N(C_i C_j) = N(C_1C_2) + N(C_1C_3) + N(C_1C_4) + N(C_2C_3) + N(C_2C_4) + N(C_3C_4)$

$N(C_1C_2) = [26 - 4 + 1 - 4 + 1]! = 20!$

$N(C_1C_3) = 0$ ($\because$ No. of permutations of distinct objects can contain both of the patterns "spin" and "path")

$N(C_1C_4) = [26 - 6 + 1]! = 21!$ ($\because$ spinet — total 6 letters; No. of permutations of distinct objects consisting of the single pattern "spinet")

$N(C_2C_3) = 0$; $N(C_2C_4) = 0$; $N(C_3C_4) = 0$

$\therefore S_2 = 20! + 21!$

we have $S_3 = \sum N(C_i C_j C_k) = N(C_1C_2C_3) + N(C_1C_2C_4) + N(C_2C_3C_4)$

$N(C_1C_2C_3) = 0$; $N(C_1C_2C_4) = 0$; $N(C_2C_3C_4) = 0$

$\therefore S_3 = 0$

we have $S_4 = \sum N(C_i C_j C_k C_l) = N(C_1C_2C_3C_4)$

$N(C_1C_2C_3C_4) = 0$

$\therefore S_4 = 0$

$\therefore \bar{N} = S_0 - S_1 + S_2 - S_3 + S_4$

$= 26! - [3(23!) + 24!] + [20! + 21!] - 0 + 0.$

Note: Floor$(x) = \lfloor x \rfloor$ is the largest integer $\leq x$.

Ex: $\lfloor 15.666 \rfloor = 15$: $\lfloor 4.566 \rfloor = 4$

$\lfloor 2.999 \rfloor = 2$

③. Determine the number of positive integers n such that $1 \leq n \leq 100$ and n is not divisible by 2, 3 or 5.

▶ Let $S = \{1, 2, 3, \dots, 100\}$

$\therefore |S| = N = S_0 = 100$

Let C_1, C_2, and C_3 be the subsets of S whose elements are divisible by 2, 3 and 5 respectively.

By the Principle Inclusion-Exclusion, we have

$\bar{N} = S_0 - S_1 + S_2 - S_3$

we have $S_1 = \sum N(C_i) = N(C_1) + N(C_2) + N(C_3)$

$N(C_1) = \left\lfloor \frac{100}{2} \right\rfloor = 50$: $N(C_2) = \left\lfloor \frac{100}{3} \right\rfloor = \lfloor 33.333 \rfloor = 33$

$N(C_3) = \left\lfloor \frac{100}{5} \right\rfloor = 20$

$\therefore S_1 = 50 + 33 + 20 = 103$

we have $S_2 = \sum N(C_i C_j) = N(C_1 C_2) + N(C_1 C_3) + N(C_2 C_3)$

$N(C_1 C_2) = \left\lfloor \frac{100}{2 \times 3} \right\rfloor = \lfloor 16.666 \rfloor = 16$

$N(C_1 C_3) = \left\lfloor \frac{100}{2 \times 5} \right\rfloor = 10$

$N(C_2 C_3) = \left\lfloor \frac{100}{3 \times 5} \right\rfloor = \lfloor 6.666 \rfloor = 6$

$\therefore S_2 = 16 + 10 + 6 = 32$

We have $S_3 = \sum N(C_i C_j C_k) = N(C_1 C_2 C_3)$

$$N(C_1 C_2 C_3) = \left\lfloor \frac{100}{2 \times 3 \times 5} \right\rfloor = \lfloor 3.333 \rfloor = 3$$

$$\bar{N} = S_0 - S_1 + S_2 - S_3 = 100 - 103 + 32 - 3 = 26.$$

(4). Six married couple are to be seated at a circular table. Find in how many ways can they arrange themselves so that no wife sits next to her husband, given that two seating arrangements are considered the same if one is a rotation of the other.

▶ The total no. of arrangements of 6 couples (i.e 12 persons) around a circular table with no restrictions is

$$|S| = N = S_0 = [12 - 1]! = 11!$$

For $1 \le i \le 6$, let C_i denote the condition where a seating arrangement has the husband and wife in a couple i seated next to each other.

By the Principal of Inclusion - Exclusion, we have

$$\bar{N} = S_0 - S_1 + S_2 - S_3 + S_4 - S_5 + S_6$$

Here $S_1 = \sum N(C_i)$

$$N(C_1) = [(12 - 2 + 1) - 1]! \times 2! = 10! \times 2!$$

$$\therefore S_1 = {}^6C_1 \times [10! \times 2!]$$

$$S_2 = \sum N(C_i C_j)$$

$$N(C_1 C_2) = [(12 - 2 + 1 - 2 + 1) - 1]! \times 2! \times 2!$$

$$= 9! \times 2^2$$

$$\therefore S_2 = {}^6C_2 \times [9! \times 2^2]$$

$$S_2 = {}^6C_2 \times 9! \times 2^2$$

$S_3 = \Sigma N(C_i C_j C_k)$

$N(C_1 C_2 C_3) = [(12-2+1-2+1-2+1) - 1]! \times 2! \times 2! \times 2!$

$= 8! \times 2^3$

$\therefore S_3 = {}^6C_3 \times [8! \times 2^3]$

Hence $S_4 = {}^6C_4 \times [7! \times 2^4]$

$S_5 = {}^6C_5 \times [6! \times 2^5]$

$S_6 = {}^6C_6 \times [5! \times 2^6]$

$\bar{N} = S_0 - S_1 + S_2 - S_3 + S_4 - S_5 + S_6$

$= 11! - [{}^6C_1 \times 10! \times 2!] + [{}^6C_2 \times 9! \times 2^2] - [{}^6C_3 \times 8! \times 2^3]$

$+ [{}^6C_4 \times 7! \times 2^4] - [{}^6C_5 \times 6! \times 2^5] + [{}^6C_6 \times 5! \times 2^6]$.

⑤. How many integers between 1 and 300 (inclusive) are (i) divisible by at least one of 5, 6, 8

(ii) divisible by none of 5, 6, 8.

▶ **Note**

$$L_m = S_m - \binom{m}{m-1} S_{m+1} + \binom{m+1}{m-1} S_{m+2} - \cdots + (-1)^{n-m} \binom{n-1}{m-1} S_n$$

Let $S = \{1, 2, 3, \cdots, 300\}$. Then $|S| = N = S_0 = 300$

Let C_1, C_2, C_3 be the subsets of S whose elements are divisible by 5, 6, and 8 respectively.

(i) The set of integers b/w 1 and 300 which are divisible by at least one of 5, 6, 8 is

$$L_1 = S_1 - {}^1C_0 S_2 + {}^2C_0 S_3 = S_1 - S_2 + S_3$$

We have $S_1 = \sum N(c_i) = N(c_1) + N(c_2) + N(c_3)$

$N(c_1) = \left\lfloor \frac{300}{5} \right\rfloor = 60$; $N(c_2) = \left\lfloor \frac{300}{6} \right\rfloor = 50$

$N(c_3) = \left\lfloor \frac{300}{8} \right\rfloor = \lfloor 37.5 \rfloor = 37$

$\therefore S_1 = 60 + 50 + 37 = 147$

We have $S_2 = \sum N(c_i c_j) = N(c_1 c_2) + N(c_1 c_3) + N(c_2 c_3)$

$N(c_1 c_2) = \left\lfloor \frac{300}{5 \times 6} \right\rfloor = 10$ $\quad N(c_1 c_3) = \left\lfloor \frac{300}{5 \times 8} \right\rfloor = \lfloor 7.5 \rfloor = 7$

$N(c_2 c_3) = \left\lfloor \frac{300}{\text{l.c.m of 6 and 8}} \right\rfloor = \left\lfloor \frac{300}{24} \right\rfloor = \lfloor 12.5 \rfloor = 12$

$\therefore S_2 = 10 + 7 + 12 = 29$

We have $S_3 = \sum N(c_i c_j c_k) = N(c_1 c_2 c_3)$

$N(c_1 c_2 c_3) = \left\lfloor \frac{300}{\text{l.c.m of 5, 6 and 8}} \right\rfloor = \left\lfloor \frac{300}{120} \right\rfloor = \lfloor 2.5 \rfloor = 2$

$\therefore S_3 = 2$

$L_1 = S_1 - S_2 + S_3 = 147 - 29 + 2 = 120$

(ii) The set of integers b/w 1 and 300 which are divisible by none of 5, 6, 8 is

$\bar{N} = S_0 - S_1 + S_2 - S_3$.

$= 300 - 147 + 29 - 2$

$= 180$.

⑥. In how many ways can one arrange the letters in the word CORRESPONDENTS so that

(i) there is no pair of consecutive identical letters?

(ii) there are exactly two pairs of consecutive identical letters?

(iii) there are at least three pairs of consecutive identical letters?

▶ Here C → 1, P → 1, D → 1, T → 1

O → 2, R → 2, E → 2, S → 2, N → 2

Let S be the set of all arrangements of 14 letters.

Then, we have $N = |S| = S_0 = \frac{14!}{2!2!2!2!2!} = \frac{14!}{(2!)^5}$

Let c_1, c_2, c_3, c_4, c_5 be the set of arrangements in which O's, R's, E's, S's and N's appear in pairs respectively.

(i) The No. of permutations, where there is no pair of consecutive identical letters.

$$\bar{N} = S_0 - S_1 + S_2 - S_3 + S_4 - S_5$$

Here $S_1 = \sum N(c_i)$

$$N(c_1) = \frac{[14-2+1]!}{[2!]^4} = \frac{13!}{[2!]^4}$$

$$\therefore S_1 = {}^5C_1 \times \frac{13!}{[2!]^4}$$

we have $S_2 = \sum N(c_i c_j)$

$$N(c_1 c_2) = \frac{[14-2+1-2+1]!}{[2!]^3} = \frac{12!}{[2!]^3}$$

$$\therefore S_2 = {}^5C_2 \times \frac{12!}{[2!]^3}$$

Hy $\quad S_3 = {}^5C_3 \times \frac{11!}{[2!]^2}$

$$S_4 = {}^5C_4 \times \frac{10!}{2!}$$

$$S_5 = {}^5C_5 \times 9!$$

$$\therefore \bar{N} = \frac{14!}{[2!]^5} - \left[{}^5C_1 \times \frac{13!}{[2!]^4}\right] + \left[{}^5C_2 \times \frac{12!}{[2!]^3}\right] - \left[{}^5C_3 \times \frac{11!}{[2!]^2}\right]$$

$$+ \left[{}^5C_4 \times \frac{10!}{2!}\right] - \left[{}^5C_5 \times 9!\right]$$

(ii). Note

$$E_m = S_m - \binom{m+1}{1} S_{m+1} + \binom{m+2}{2} S_{m+2} - \cdots + (-1)^{n-m} \binom{n}{n-m} S_n$$

The number of permutations where there are exactly two pairs of consecutive identical letters is

$$E_2 = S_2 - {}^3C_1 S_3 + {}^4C_2 S_4 - {}^5C_3 S_5$$

$$= {}^5C_2 \times \frac{12!}{[2!]^3} - \left({}^3C_1 \times {}^5C_3 \times \frac{11!}{[2!]^2}\right) + \left[{}^4C_2 \times {}^5C_4 \times \frac{10!}{2!}\right]$$

$$- \left[{}^5C_3 \times {}^5C_5 \times 9!\right]$$

(iii). Finally, the number of permutations where there are at least three pairs of consecutive identical letters is

$$L_3 = S_3 - {}^3C_2 S_4 + {}^4C_2 S_5$$

$$= {}^5C_3 \times \frac{11!}{[2!]^2} - \left[{}^3C_2 \times {}^5C_4 \times \frac{10!}{2!}\right] + \left[{}^4C_2 \times {}^5C_5 \times 9!\right]$$

(7). Determine in how many ways can the letters in the word ARRANGEMENT be arranged so that

(i) there are exactly two pairs of consecutive identical letters

(ii) at least two pairs of consecutive identical letters?

(H.W) ~~A→2, R→2, N→2, E→2, G→1, M→1, T→1~~

A→2, R→2, N→2, E→2, G→1, M→1, T→1

$$|S| = N = S_0 = \frac{11!}{[2!]^4}$$

Let c_1, c_2, c_3, c_4 denote the presence of consecutive A's, R's, N's, and E's respectively in the arrangement.

$$N(c_1) = \frac{[11-2+1]!}{(2!)^3} = \frac{10!}{[2!]^3} \quad \therefore S_1 = {}^4C_1 \times \frac{10!}{[2!]^3}$$

$$S_2 = {}^4C_2 \times \frac{9!}{[2!]^2} \quad \therefore S_3 = {}^4C_3 \times \frac{8!}{2!} \quad \therefore S_4 = {}^4C_4 \times 7!$$

(i) $E_2 = S_2 - {}^3C_1 S_3 + {}^4C_2 S_4$

(ii) $L_2 = S_2 - {}^2C_1 S_3 + {}^3C_2 S_4$